

Effects of comprehensive lifestyle modification on diet, weight, physical fitness, and blood pressure control: 18-month results of a randomized trial.



BACKGROUND: The main 6-month results from the PREMIER trial showed that comprehensive behavioral intervention programs improve lifestyle behaviors and lower blood pressure. **OBJECTIVE:** To compare the 18-month effects of 2 multicomponent behavioral interventions versus advice only on hypertension status, lifestyle changes, and blood pressure. **DESIGN:** Multicenter, 3-arm, randomized trial conducted from January 2000 through November 2002. **SETTING:** 4 clinical centers and a coordinating center. **PATIENTS:** 810 adult volunteers with prehypertension or stage 1 hypertension (systolic blood pressure, 120 to 159 mm Hg; diastolic blood pressure, 80 to 95 mm Hg). **INTERVENTIONS:** A multicomponent behavioral intervention that implemented long-established recommendations ("established"); a multicomponent behavioral intervention that implemented the established recommendations plus the Dietary Approaches to Stop Hypertension (DASH) diet ("established plus DASH"); and advice only. **MEASUREMENTS:** Lifestyle variables and blood pressure status. Follow-up for blood pressure measurement at 18 months was 94%. **RESULTS:** Compared with advice only, both behavioral interventions statistically significantly reduced weight, fat intake, and sodium intake. The established plus DASH intervention also statistically significantly increased fruit, vegetable, dairy, fiber, and mineral intakes. Relative to the advice only group, the odds ratios for hypertension at 18 months were 0.83 (95% CI, 0.67 to 1.04) for the established group and 0.77 (CI, 0.62 to 0.97) for the established plus DASH group. Although reductions in absolute blood pressure at 18 months were greater for participants in the established and the established plus DASH groups than for the advice only group, the differences were not statistically significant. **LIMITATIONS:** The exclusion criteria and the volunteer nature of this cohort may limit generalizability. Although blood pressure is a well-accepted risk factor for cardiovascular disease, the authors were not able to assess intervention effects on clinical cardiovascular events in this limited time and with this sample size. **CONCLUSIONS:** Over 18 months, persons with prehypertension and stage 1 hypertension can sustain multiple lifestyle modifications that improve control of blood pressure and could reduce the risk for chronic disease.